Matheism: Why Reality Is Mathematical and What Follows for Physics

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Abstract

Anything that affects the world must have properties and relations. Matheism argues that these supply mathematical structure, leaving no separate substance to explain. This paper connects that claim to a physical proposal: a short rule can generate an elaborate world, including its observers. Under a fixed binary description measure, every extra bit halves a description's initial weight. The comparison counts laws, starting conditions and predictions together, so hidden complexity cannot be moved out of sight. The paper connects this economy to quantum evolution without collapse and to the proposed development of time's arrow, complexity and minds. A mechanism of consciousness would itself add structure; it would not escape the argument. The philosophical claim concerns what reality is. The physical proposal asks which complete mathematical account explains our observations most economically.

Keywords: mathematical monism, rule-random dichotomy, Kolmogorov complexity, structural realism, Simplicity Dominance Principle

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Matheism is mathematical monism, expressed here as *Deus sive Mathematica*: God, or mathematics. The formula names the proposed identity of reality and mathematical structure, not a creator using equations or written symbols producing matter. Spinoza is a predecessor in substance monism and necessity, not a prior advocate of this exact identity. His geometric presentation does not itself establish a mathematical ontology. Causal powers and change belong within the proposed structure, not outside it. See Spinoza, *Ethics* I.14, I.29, I.33 and I.34.

1. The Proof

1.1 Existence requires determinacy

For something to exist, it must have a nature. It must differ from nothing in some respect. A property, a possible change, or an effect on something else supplies relations that can be stated mathematically. The question is what, if anything, remains when all those features are specified. Matheism's answer is that the complete structure is the thing itself.

Properties do not float free. Each is defined by its connections to others. Mass relates to energy through $E=mc^2$. Position relates to velocity through derivative

relationships. A property's identity is constituted by how it relates to other properties within the overall structure. Properties form a relational structure.

Physics has progressively confirmed this. What seemed like solid particles became excitations in fields. Fields became representations of symmetry groups. Forces became geometric curvatures in spacetime. At each level, the concrete substance dissolved into relational structure. The electron is not a tiny ball with properties attached. It is a pattern of relationships: how it couples to the electromagnetic field, how it responds to measurements, how it relates to other fermions through the Pauli exclusion principle. Remove these relations and nothing remains. Structure is the substance. The Standard Model is purely relational: gauge fields, coupling constants, symmetry groups. No underlying stuff.

1.2 The dichotomy

Determinate features are either ruled or random. This is made precise by algorithmic information theory. A pattern is a structure whose Kolmogorov complexity K(S) is strictly less than its length: there exists a rule simpler than the data that generates it. The laws of physics, the digits of pi, fractal coastlines, deterministic chaos are all patterns. Short programs generating vast complexity.

Randomness is the complement: K(S) is approximately equal to the length of S. The sequence is algorithmically incompressible. No shorter description exists. Crucially, randomness is not the absence of mathematics. Kolmogorov complexity is itself a mathematical function. The statement "this sequence is random" is a mathematical claim about a mathematical object. Even structurelessness is structure, since it is defined and measured mathematically.

Since K is a well-defined function mapping any finite string to an integer, both categories are strictly mathematical objects. Every finite sequence is either compressible or incompressible. There is no third option. The dichotomy is exhaustive. This is not a tautology achieved by defining mathematics broadly. The claim that randomness is mathematical is a synthetic discovery of twentieth-century algorithmic information theory, not a trivial restatement.

Any proposed non-mathematical substance must have determinate features, otherwise it is nothing. Those features are ruled or random. Both are mathematical. The proposed substance is mathematical. The escape routes all lead back. Any objection must itself be structured, which is to say: any objection uses structure to deny structure.

Stochastic laws are mathematically coherent. Identical preparation conditions can determine a probability distribution without determining one outcome. Rejecting this possibility requires an additional physical or metaphysical argument; it does not follow from the Rule-Random Dichotomy. Algorithmic randomness of a sequence, stochasticity of a model and unpredictability for an observer are distinct concepts.

1.3 Interaction collapses dualism

An interaction requires a determinate relation between its relata. A theory in which mind changes bodily events must therefore specify how that relation works. This constrains an appeal to a wholly nonstructural soul. It does not, without further premises, show that everything related by one law is the same substance. The stronger identity claim depends on the argument in Section 1.4 that relational structure exhausts the relata's properties and causal powers.

Descartes associated mind and body interaction with the pineal gland. In the preface to Ethics V, Spinoza challenges the account of how an unextended mind could determine bodily motion. His alternative is not a different anatomical bridge: in Ethics II.7, scholium, one individual is understood through thought and extension. This supports the historical comparison with mathematical monism while leaving the further identification with mathematical structure to the present argument. The gland objection concerns a missing account of interaction; it is not a measure theoretic proof that a point cannot exert force. Spinoza, Ethics.

The argument applies recursively. Suppose someone posits a non-mathematical property, call it Q, that interacts with mathematical structure. For Q to interact with structure, Q must have features that structure can couple to. Those features are determinate (else the coupling is undefined). Determinate features are ruled or random. Both are mathematical. Q is mathematical. The dualism was apparent. Every proposed non-mathematical layer reduces to mathematics on contact with anything that is mathematical, which is everything.

1.4 The identity argument

Physical reality and its isomorphic mathematical structure share all relational properties. What distinguishes them? Not their relationships: identical by definition of isomorphism. Not their causal powers: causal powers supervene on relational structure. Not their observable properties: all observable properties reduce to relational structure. Color is wavelength. Hardness is bonding strength. Temperature is molecular motion.

The only remaining candidate is non-relational substrate: some primitive stuff that bears properties without being constituted by them. But physics has eliminated it. Quantum field theory contains no objects with intrinsic properties independent of their role in relational structure. And substrate does no explanatory work. Two systems with identical relational structure but hypothetically different substrates behave identically in every observable way. The substrate is explanatorily idle.

Any proposed substrate must itself have determinate features, else it is nothing. Those features are ruled or random. Both are mathematical. Substrate is structure. The regress terminates. By the Identity of Indiscernibles: if two things share all properties, they are identical. Physical reality and its isomorphic

mathematical structure share all relational properties. They are not merely similar. They are identical. The map is the territory.

Consider chess implemented in carved wood, in computer memory, or as pure mathematical abstraction. No chess property differs across implementations. No legal move, no winning strategy, no checkmate changes. The game is the relational structure. The wood is merely one access-mode. Burning the wood destroys a token, not the type. The distinction marks a difference in our access, not in the object accessed.

1.5 Transcendental constraint

Reliable observation requires structured relations: a record must track what it records, and a learned cue must bear on the event it predicts. This is a necessary condition of reliable observation. It does not require every observed object, or the observer's complete stored data, to be compressible. An observer can store an incompressible string. Stochastic laws also have mathematical structure. The constraint concerns the relations supporting observation, rather than a universal ban on randomness.

1.6 Mathematical structure is necessary

Mathematical structure is uncreated and necessary. No agent makes 2+2=4, and none could make it otherwise. No conceivable process could make 2+2 cease to equal 4. The rules are the things. The question "why is there something rather than nothing?" dissolves at necessity rather than brute fact. Mathematical structure cannot fail to exist. Asking what caused it commits a category error. Causation is a relation within structure. Math is what causation is.

Only consistent mathematics forms structure. Inconsistent statements can be written but not constructed. "The set of all sets that do not contain themselves" is syntactically valid and structurally void. You can label a contradiction. You cannot build one. Consistency is the filter on what exists.

2. All Consistent Structures Exist

Universal realization is a further ontological commitment. If every consistent structure is physically realized, consistent selection rules are included. The reverse inference does not follow: the consistency of a filter, including the filter that excludes nothing, does not establish its physical realization without assuming the ensemble being argued for. Mathematical monism motivates the proposal but does not by itself prove that every mathematical possibility is physical.

The distinction from Tegmark is the proposed route to structural identity. Universal realization remains a further thesis to defend, rather than a consequence licensed by the null filter alone.

Under this universal realization thesis, every consistent law structure exists in its own right. That does not place every structure inside our particular universe. The ontological argument concerns what existence is; the measure question concerns which structures and observations receive weight. Keeping those questions distinct allows each part of the Matheist position to receive its own defence.

3. The Measure

Fix a description language and a class of complete candidate models. With assigned code weights w(H) proportional to $2^{\wedge}(-\mathrm{L}(\mathrm{H}))$, a genuine difference of ΔL bits gives a prior ratio $2^{\wedge}\Delta\mathrm{L}$. Ten bits gives exactly 1,024. Posterior odds additionally multiply by the evidence likelihood ratio. Summing over all programs producing an output is different from assigning only its shortest program weight. Neither construction automatically defines a normalized measure over every mathematical structure or every observer.

Algorithmic probability was developed for induction. Its use as an ontological or observer measure is a proposed extension. The coding domain, multiplicities and conditioning rule must be explicit before its physical predictions can be assessed.

The simplest programs may generate no observers. Conditioning on the evidence therefore changes the comparison. A longer candidate or a large class of candidates can outweigh a shorter one after likelihoods and observer weights are included. No verified total bit count of observed physics is assumed here.

A refinement worth naming. Our universe's measured parameters (the nineteen coupling constants, masses, and mixing angles of the Standard Model) are commonly counted as independent specifications adding to the description length. They may not be. They may be addresses within a simpler generating space, the way a city is a region on a map: the city is in the map already, and the cost is the map's, not the city's. Under this branch reading, the K of our universe is the K of the generating rule plus a small address component, not the full parameter enumeration. The exactness of the parameters (every electron with the same mass to twelve decimal places) is evidence: exact values are properties of mathematical regions, not emergent averages from sub-structure. The branch reading sharpens SDP without altering its conclusions: our universe is even more favored than the conservative bit-count suggests, and the comparison with the string landscape becomes more decisive, not less.

4. Five Consequences

4.1 What the simplicity prior establishes

Under a specified code prior, shorter descriptions receive greater individual prior weight. This gives a precise version of a simplicity preference. It does not guarantee that the shortest candidate is true or best supported after evidence.

The proposed ontological extension asks whether this coding preference also explains observed physical economy. Success requires predictive comparisons rather than treating the chosen prior as an observed distribution of existence.

4.2 The arrow of time

The proposed arrow concerns the production of information. In a history with a first event and no last, forward generation can begin with supplied information. A reverse dependency chain in which every step waits for a later step cannot start without an independently supplied input. This excludes that reverse construction. Connecting it to experienced time requires showing why experience needs such production, rather than the relevant relations simply existing together in a mathematical block.

If a history starts without the organization its later observers require, that organization must develop. The laws themselves can remain simple. In biological histories like ours, available energy supports that development while total entropy increases. Neither organization nor entropy must increase at every step.

Coordinated convergences in reversed splashes may depend on details a local learner cannot sense. If the same available evidence hides situations requiring incompatible responses, no decision based on that evidence can guarantee success in both. Later observations may explain the event, too late for the earlier choice. Repeated dependence on such inaccessible information could constrain the evolution of reliable learners. A directional preference must survive a comparison of observers and environments together: full reversal also transforms the records.

Start, hypothetically, with neither temporal direction favored: a coin flip. The proposed arguments then favor forward development through rooted information production, the development of initially absent observer organization and reliable evidence available before action. Even if reversal is possible, it could receive less probability weight under the stated measure. Establishing that preference requires showing that additional specification or restrictions on viable observer histories actually reduce their total weight, including comparison with a fully reversed observer and environment. The arguments share assumptions and cannot be counted as independent probability gains.

A demonstrated extra description cost would supply another preference. Under the same weighting by $2^{(-L)}$, where L is the complete description length in bits, five extra bits reduce individual weight by a factor of 32. A reversal instruction added to one program does not prove that every reverse description needs it. A compact rule can specify coordinated events.

A simplicity prior can compare complete histories, including their starting conditions. Showing that developmental histories dominate after conditioning on observers still requires that comparison. A short description alone does not guarantee a low-entropy beginning.

4.3 What observer selection explains

Observer selection constrains which constants we can observe, but a quantitative fine tuning explanation also needs a candidate space, measure and observer weighting. Simplicity can enter that comparison; it does not prove that our constants are the shortest compatible values.

The account seeks a common explanation of laws and observer conditions. Its success would be a quantitative distribution of observations, not the mere availability of an observer permitting structure.

4.4 The hard problem of consciousness

The hard problem asks why physical processing is experienced. The author's explanation constraint is that any explanation must specify determinate properties and relations. If a critic excludes structure from constituting experience in advance, adding further structured explanations cannot meet that demand. This challenges a stipulated separation, not the legitimacy of asking what explains experience. Chalmers's question does not require unstructured language or an unstructured theory. The identity defended here is the further thesis that a complete constitutive structure leaves no additional experiential ingredient.

The proposed account identifies sentience with felt transitions in modeling systems and reflective consciousness with transitions in systems that model their own activity. The threshold and the mapping from organization to specific experience remain research questions. If complete constitutive structural identity is established, an experiential difference requires a structural difference. Merely assuming that conclusion would not answer the zombie objection. Neither generic recursion nor the structured form of an explanation by itself establishes the phenomenal identity claimed here.

Schopenhauer explains particular acts through motives and character while placing will as such outside the demand for a further ground. The present objection targets that stopping point, not an absence of any account of action. Likewise, assigning experience a fundamental status does not alone specify the organization of subjects or their desires. Matheism proposes to explain both within one structural ontology. Its account of necessary existence and its derivation of particular minds are separate obligations; success at the first does not automatically complete the second.

4.5 Many-to-one as structural root

The Principle of Least Action, the thermodynamic arrow and emergence suggest a common research programme connecting microscopic structure to observable physics. Their mechanisms must be distinguished. Stationary phase can recover classical action equations in suitable regimes. Coarse graining groups microstates. Boundary conditions and statistical assumptions can support an entropy gradient. None of these results follows from noninjectivity alone.

A short discrete rule is worth searching for when it can recover the required geometry, dynamics and observations. Its effective theory must be derived. Noether's theorem operates within a variational framework; it does not prove that all observer supporting structures possess one.

I argue for a discrete physical account of extension. A sum of singleton lengths is zero, including the generalized nonnegative sum over an uncountable family. Analysis does not identify that sum with interval length. In the standard real line, order, metric and normalized measure provide the structure used to calculate interval properties; order alone does not specify their size.

Representation is broader than approximation. Discrete and continuous models can agree exactly on a specified observable while differing on others. An exact mean neither supplies occupied positions between the discrete states nor identifies their full distributions. The physical task is to recover observed distances and dynamics from specified discrete relations.

A bearer requirement asks what accounts for extension. It does not by itself exclude a continuum, whose metric relations already encode distance. Discrete models must also account for their scales or show how those scales emerge.

5. Falsification

The position is falsifiable through its consequences. If the final theory of physics requires irreducibly high complexity, SDP is challenged. A counterexample to a claimed universal variational emergence would reject that conjecture. If a force or phenomenon is discovered with no geometric or structural origin, mathematical monism faces an anomaly. If consciousness is demonstrated to depend on non-structural properties, something beyond substrate-independence, the hard-problem dissolution fails.

These are constraints, not escape hatches. Call the conclusion tautological: so is 2+2=4. A tautology on reality itself is the deepest possible truth. It is also the most exposed: any counterexample to the dichotomy refutes the entire edifice.

Conclusion

Matheism offers one account of laws, matter and minds: they are mathematical structure. Its central argument asks what a further substance could contribute beyond a complete account of its properties and relations. Quantum mechanisms for consciousness, if established, would remain within that account.

The physical work is to identify the structure of our universe. A shorter complete description receives greater initial weight under the stated simplicity rule. A history starting without the organization required by its later observers must develop it. Explaining our particular laws, quantum observations and direction of time requires showing how those results fit together in a complete model.

References

Chalmers, D. J. (1996). *The Conscious Mind: In Search of a Fundamental Theory*. Oxford University Press.

Schmidhuber, J. (2000). Algorithmic theories of everything. *arXiv:quant-ph/0011122*.

Solomonoff, R. J. (1964). A formal theory of inductive inference. *Information and Control*, 7(1), 1-22.

Tegmark, M. (2008). The mathematical universe. *Foundations of Physics*, 38(2), 101-150.